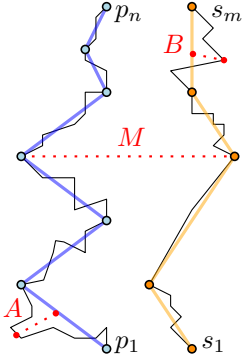

Intuition of slack(v):

Can v reach the other curve, using the purple leash?

If slack(v) is negative:

Add v to $\overline{\pi}$ (or $\overline{\sigma}$)